Regression Diagnostics

First we set the seed for the random number generator so that every time we simulate the data we get the same results.

```
set.seed(0)
```

We will simulate eight data sets.

```
n<-100                                          #sample size=100
sigma<-5                                        #standard deviation=5
x<-1:n                                          #x values are 1,2,...,n

e1<-rnorm(n,0,sigma)
y1<-x+e1                                        #1st model, no assumption violated

y2<-x+0.02*x^2+e1                               #2nd model, quadratic model

e3<-rnorm(n,0,(1:n)/n*sigma*3)
y3<-x+e3                                        #3rd model, variance increases with x

e4<-rnorm(n,0,sigma)
e4[n/2]<-e4[n/2]+50
y4<-x+e4                                        #4th model, an outlier

e5<-rt(n,2)*sigma
y5<-x+e5                                        #5th model, t error

e6<-arima.sim(model=list(ar=0.7),n)*sigma
y6<-x+e6                                        #6th model, postive correlation, AR(1)

e7<-arima.sim(model=list(ar=-0.7),n)*sigma
y7<-x+e7                                        #7th model, negative correlation, AR(1)

x2<-rnorm(n,0,5)#rep(0,n)
y8<-x+x2+e1
#8th model, lurking variable
```

Now we fit eight simple linear regression model.

```
m1<-lm(y1~x)
m2<-lm(y2~x)
m3<-lm(y3~x)
m4<-lm(y4~x)
m5<-lm(y5~x)
m6<-lm(y6~x)
m7<-lm(y7~x)
m8<-lm(y8~x)
```

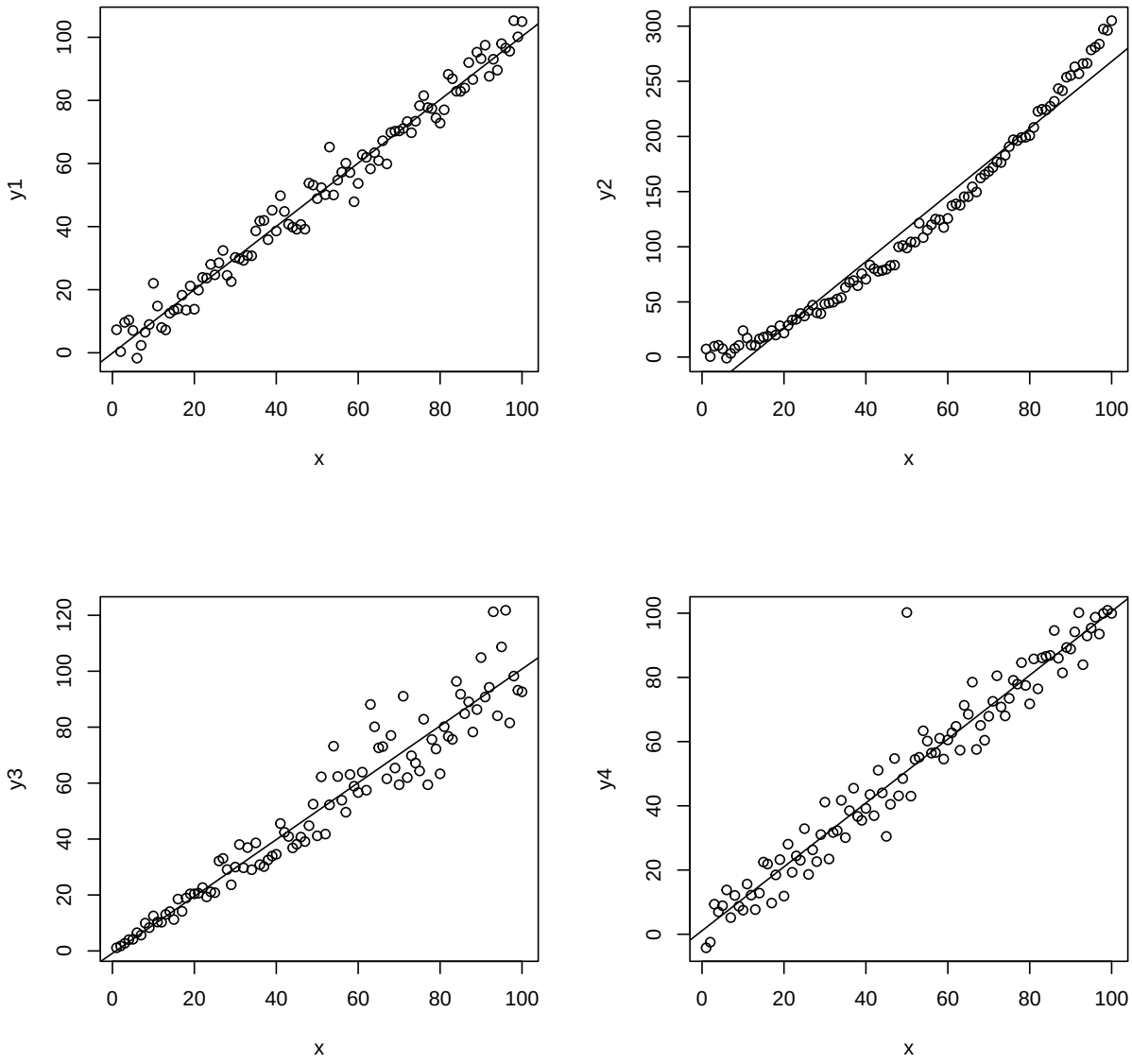
Scatterplots with the sample regression lines, first four models.

```
par(mfrow=c(2,2))
plot(x,y1)
abline(m1)

plot(x,y2)
abline(m2)

plot(x,y3)
abline(m3)

plot(x,y4)
abline(m4)
```



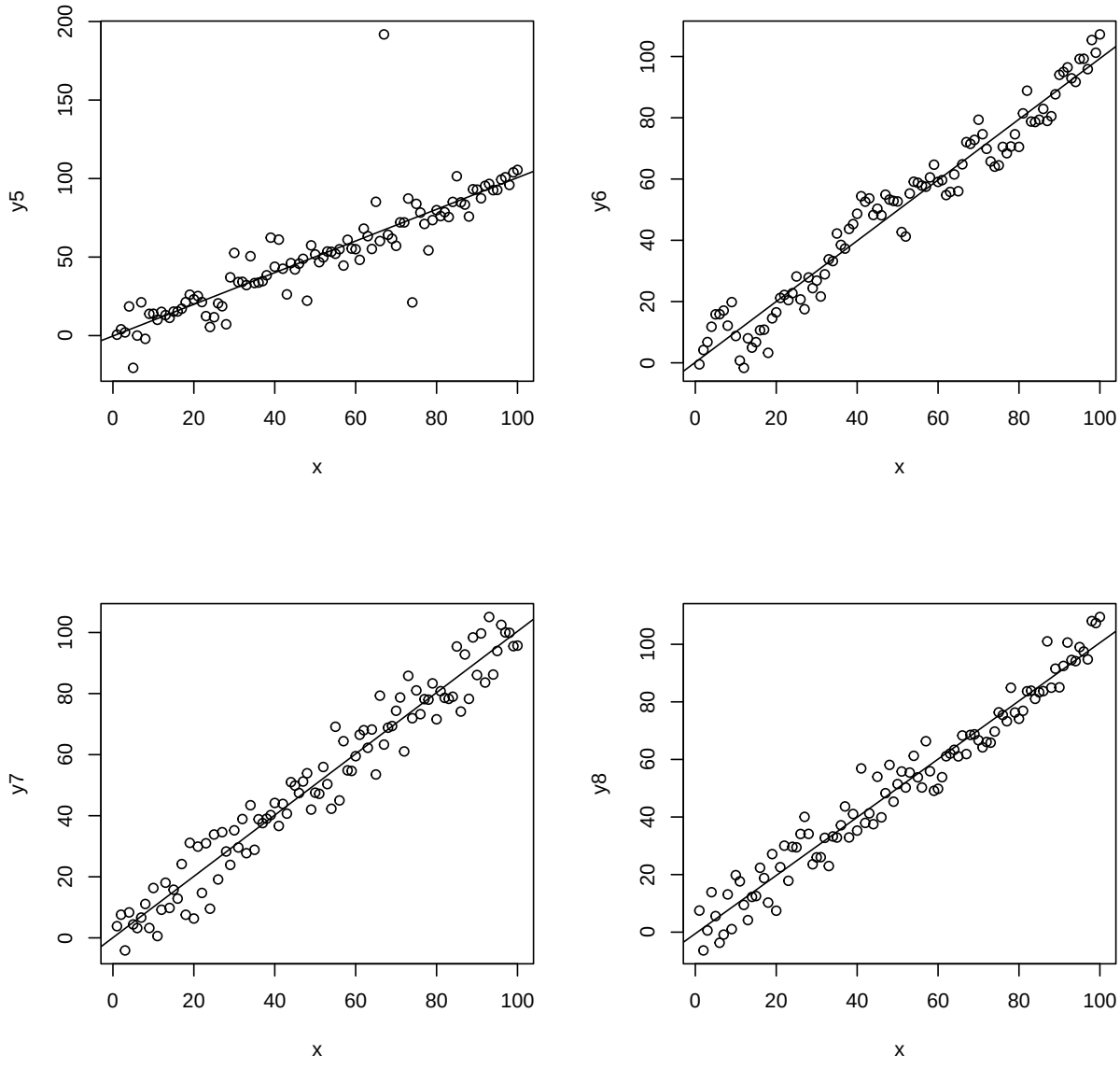
Scatterplots with the sample regression lines, second four models.

```
par(mfrow=c(2,2))
plot(x,y5)
abline(m5)

plot(x,y6)
abline(m6)

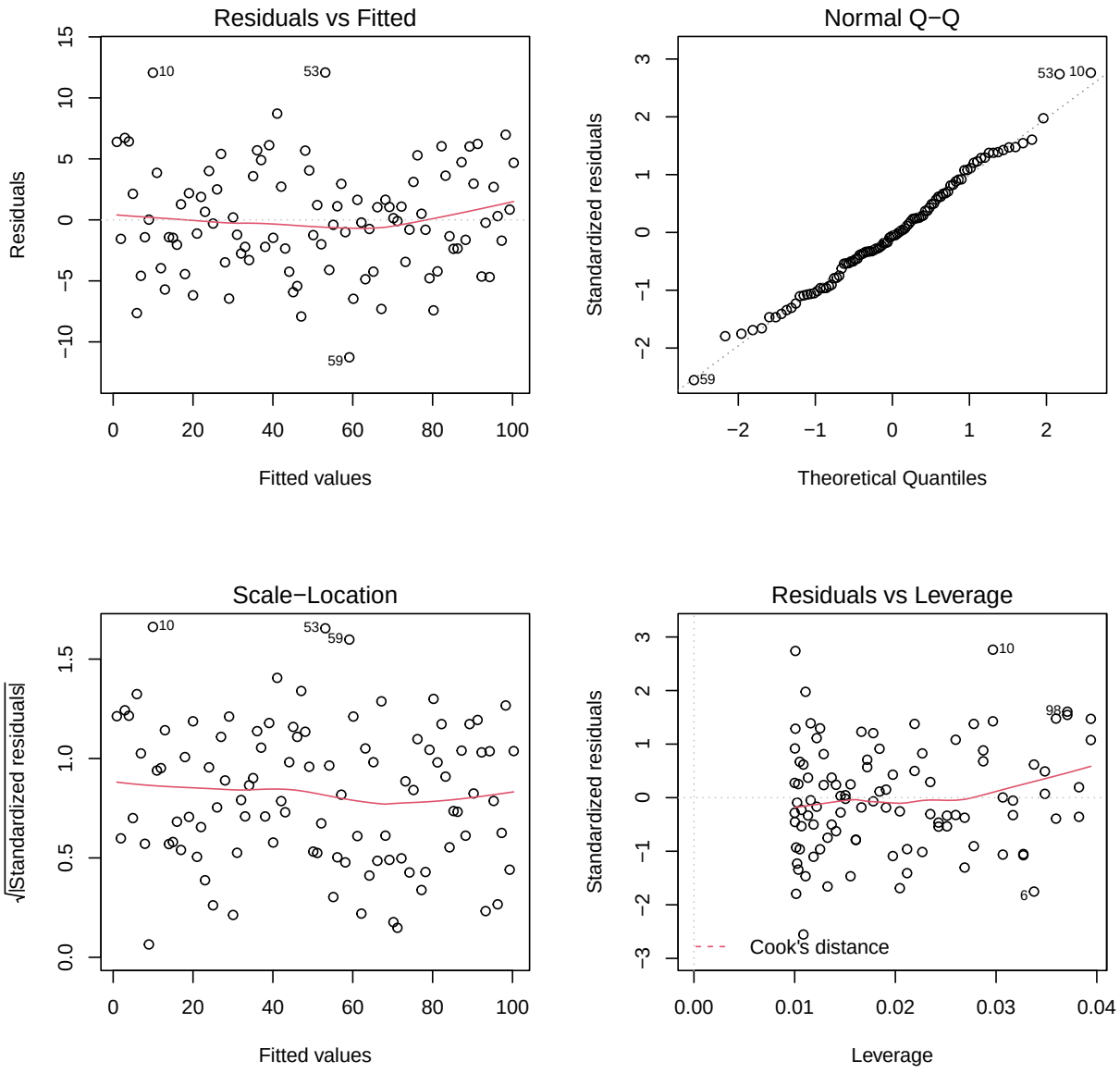
plot(x,y7)
abline(m7)

plot(x,y8)
abline(m8)
```

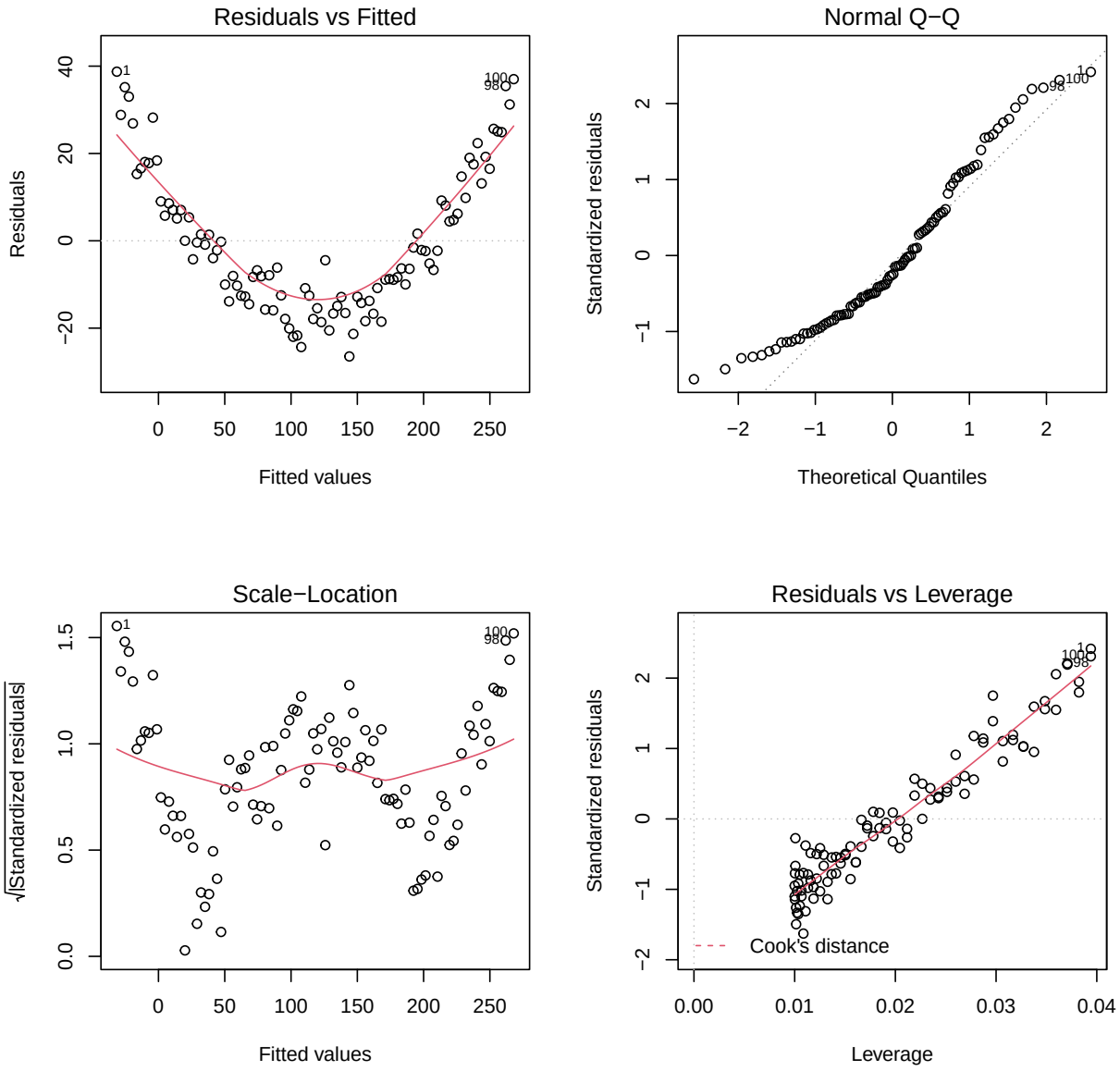


Obtain the diagnostic plots for each model.

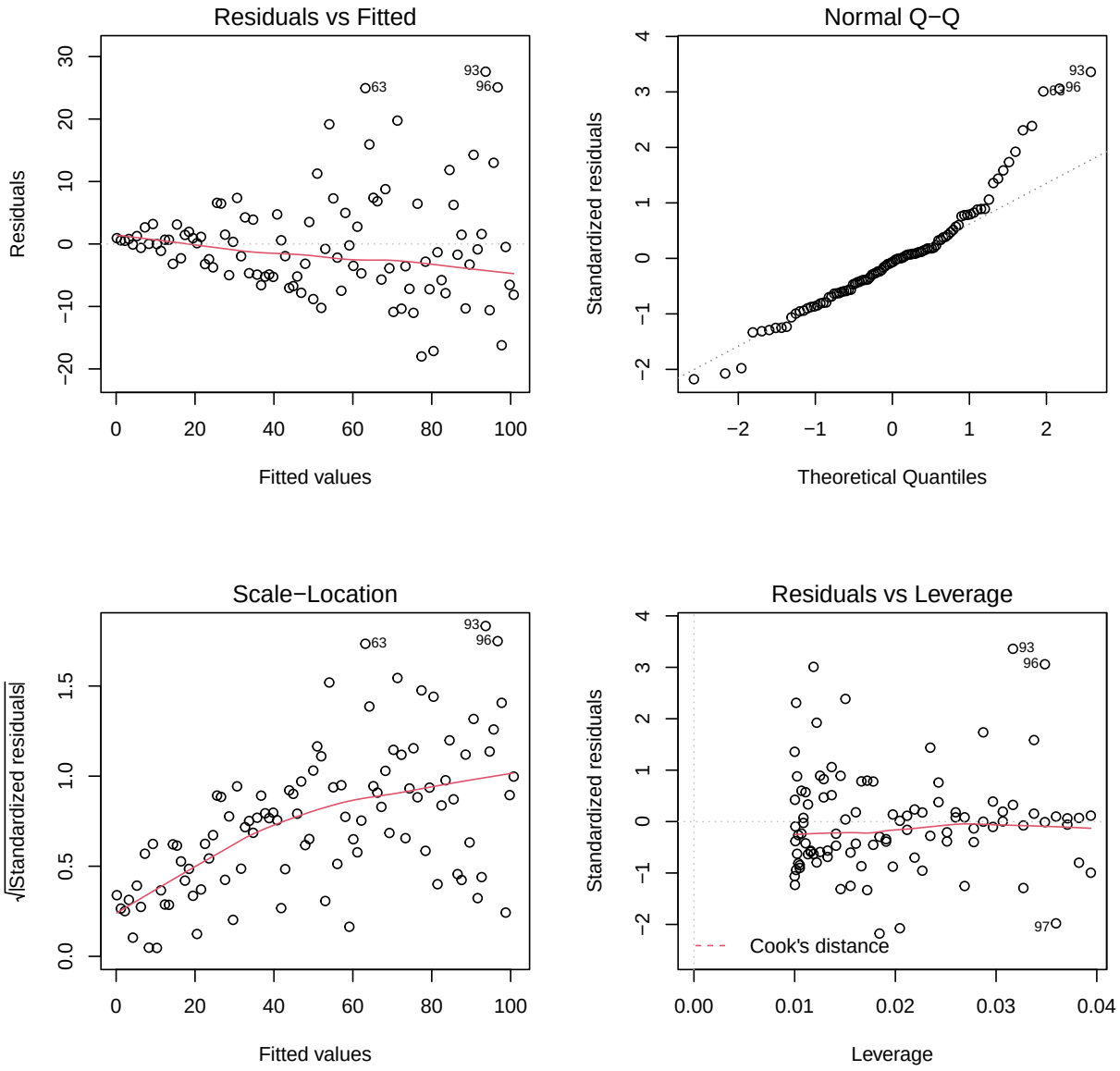
```
par(mfrow=c(2,2))
plot(m1)
```



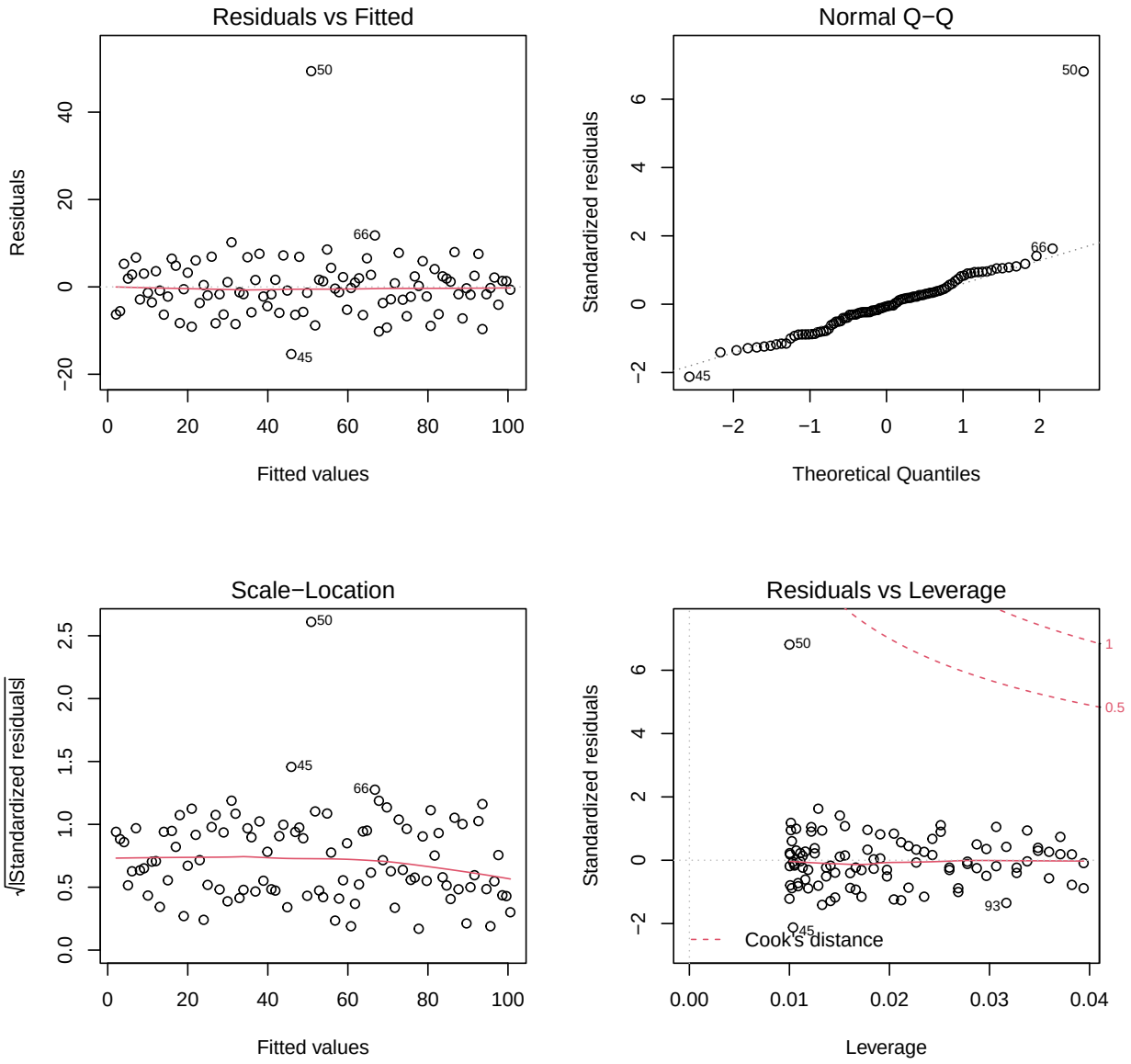
```
par(mfrow=c(2,2))
plot(m2)
```



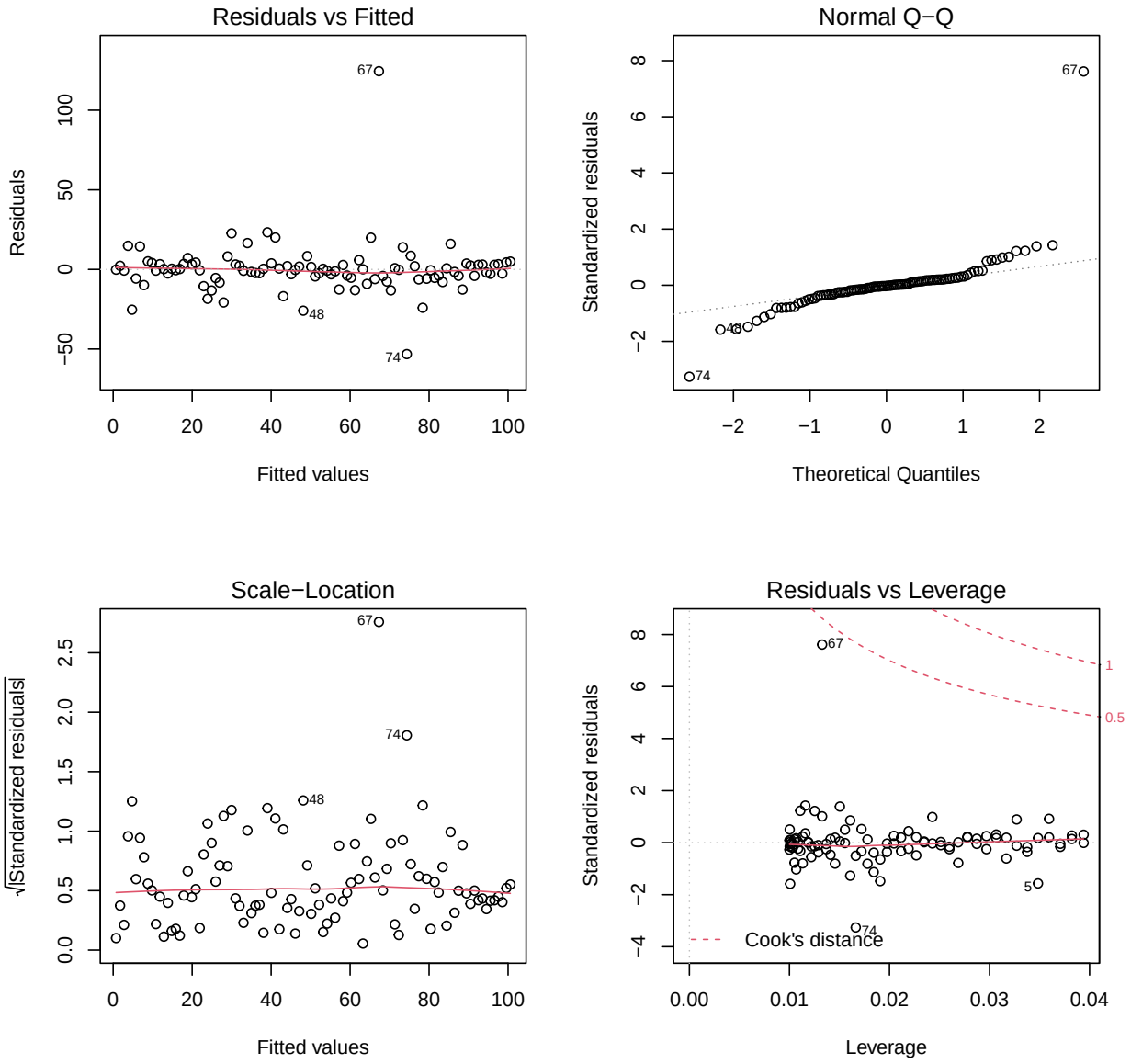
```
par(mfrow=c(2,2))
plot(m3)
```



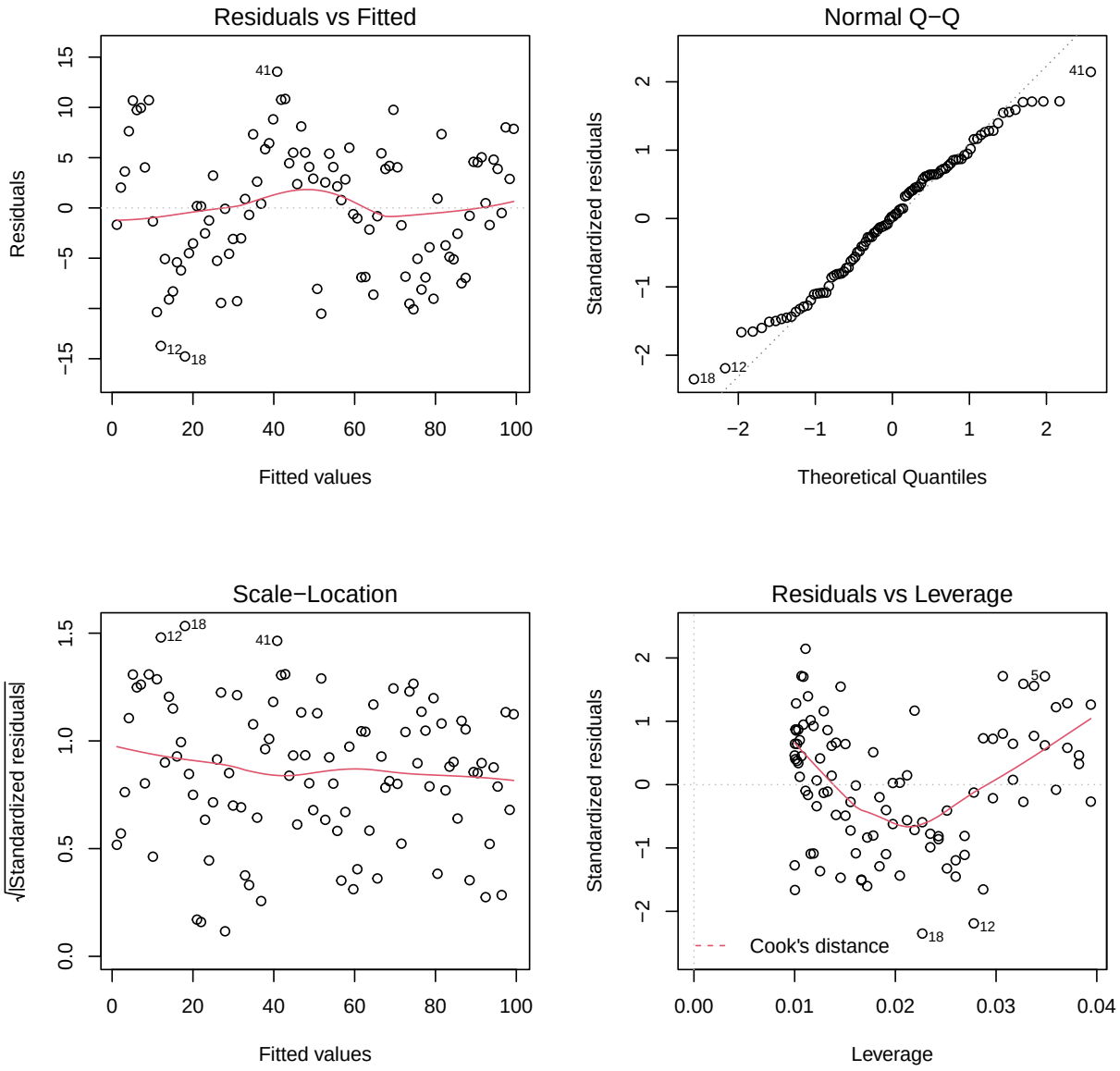
```
par(mfrow=c(2,2))
plot(m4)
```



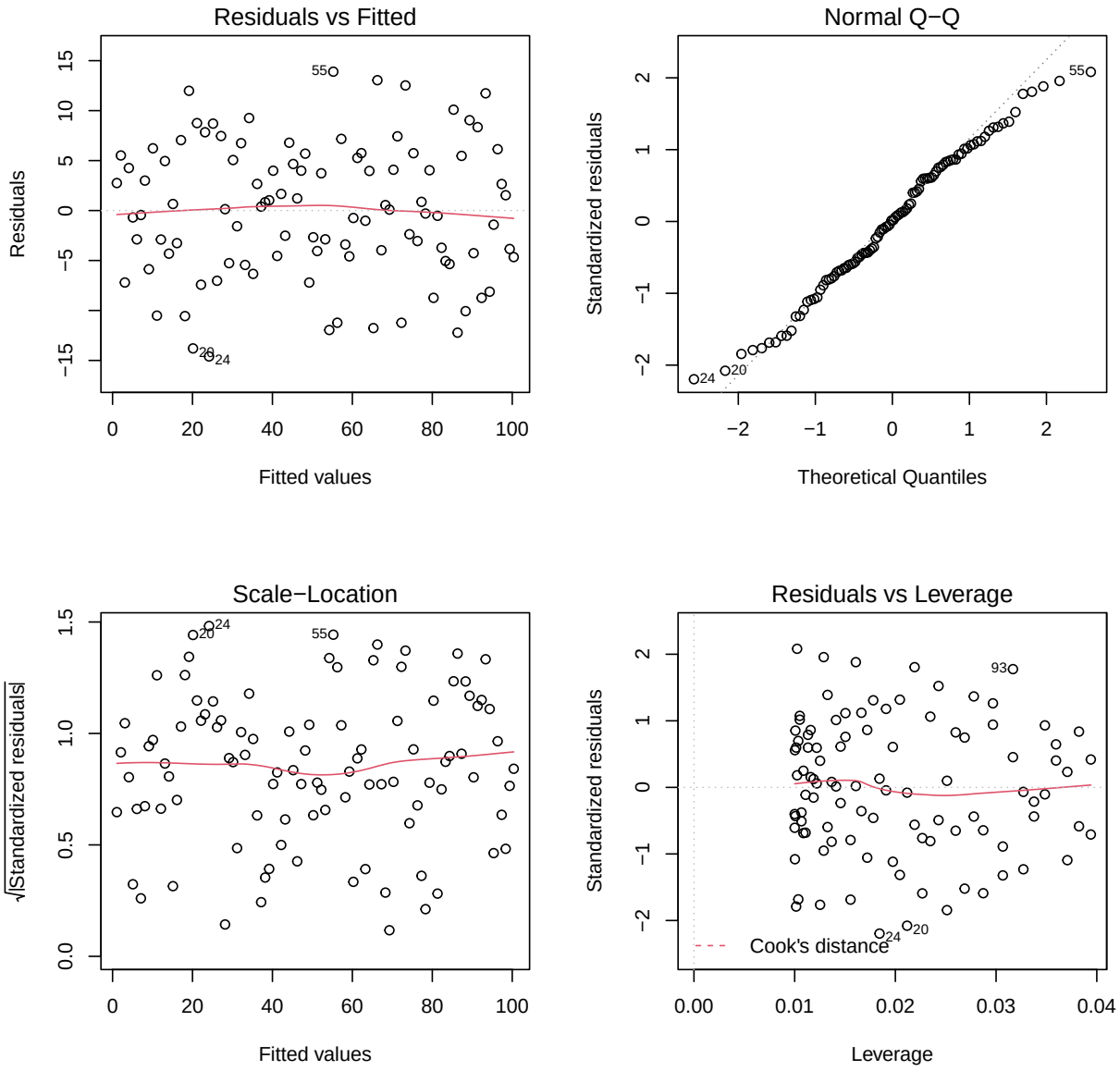
```
par(mfrow=c(2,2))
plot(m5)
```



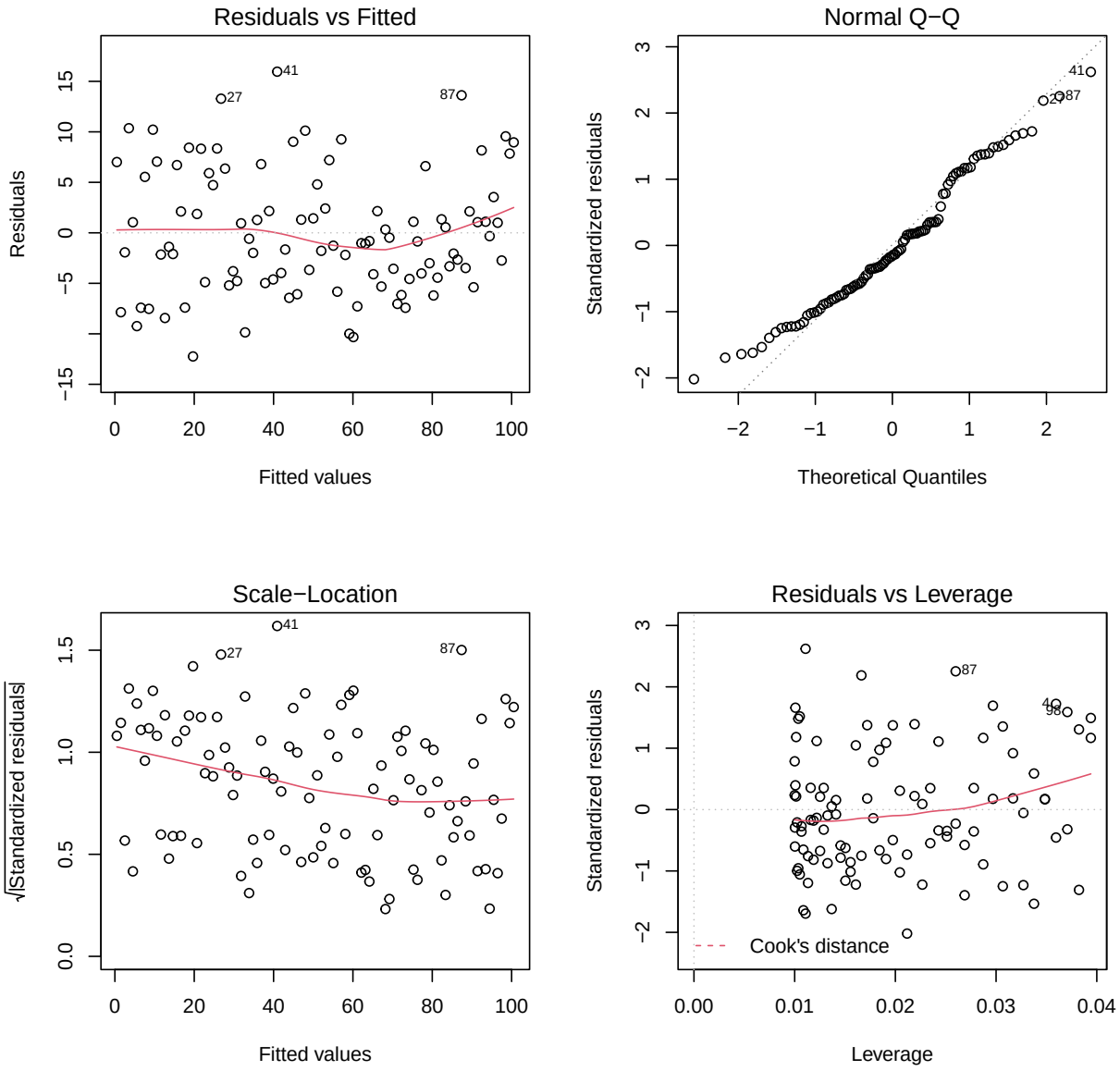
```
par(mfrow=c(2,2))
plot(m6)
```



```
par(mfrow=c(2,2))
plot(m7)
```



```
par(mfrow=c(2,2))
plot(m8)
```

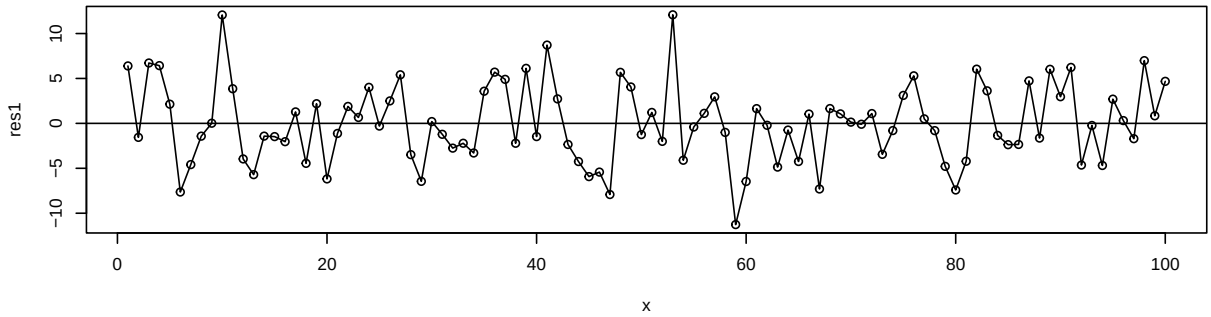


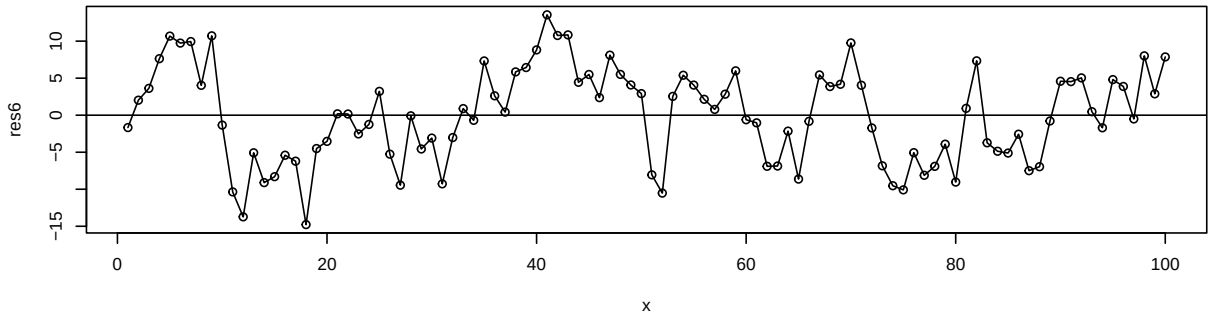
Compare the residual plots for models 1, 6 and 7.

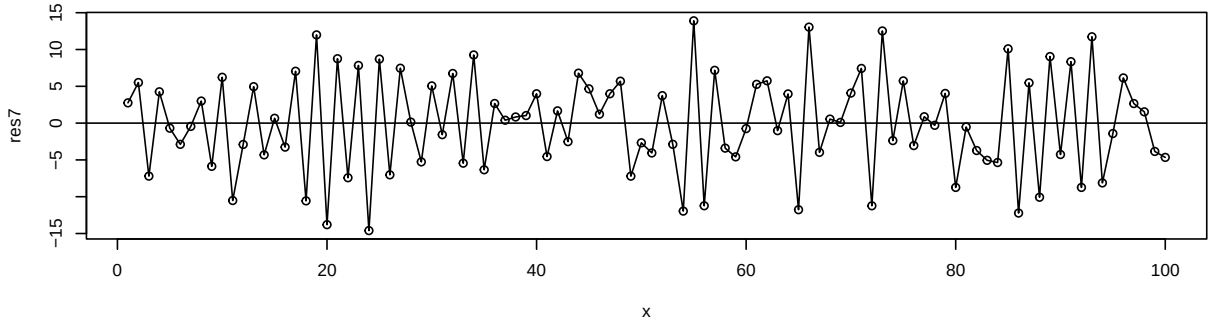
```
par(mfrow=c(3,1))
res1<-m1$residuals
plot(x,res1)
abline(h=0)
lines(x,res1)

res6<-m6$residuals
plot(x,res6)
abline(h=0)
lines(x,res6)

res7<-m7$residuals
plot(x,res7)
abline(h=0)
lines(x,res7)
```


res1
10
5
0
−5
−10
0
20
40
60
80
100
x


res6
10
5
0
−5
−15
0
20
40
60
80
100
x


res7
15
10
5
0
−5
−15
0
20
40
60
80
100
x

Compare the residual plots for models 1 and 8.

```
par(mfrow=c(2,1))

plot(x2,res1)
abline(h=0)

res8<-m8$residuals
plot(x2,res8)
abline(h=0)
```

